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Inventor(s)	INNOCENT; Manuel H.

SINGLE-ENDED ANALOG-TO-DIGITAL CONVERTER INPUT STAGE WITH TRANSISTOR-BASED CAPACITANCE

Abstract

An illustrative input stage for an analog-to-digital converter circuit is described herein. The input stage may include a first transistor of a first conductivity type connected between a first node and a second node; a second transistor of a second conductivity type connected between the first node and a third node; and a set of switches. The set of switches may be configured, when manipulated by a controller, to connect the first node to either an input node or a first output node, the second node to either a first reference node or a second output node, and the third node to either a second reference node or the second output node. Corresponding analog-to-digital conversion circuits and procedures making use of this input stage, as well as corresponding systems, circuitry, and methods, are also disclosed.

Inventors: **INNOCENT; Manuel H. (Wezemaal, BE)**

Applicant: **SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC** (Scottsdale, AZ)

Family ID: **96738302**

Assignee: **SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC (Scottsdale, AZ)**

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Background/Summary

TECHNICAL FIELD

[0001] This description relates to analog-to-digital conversion circuitry such as may be implemented within image sensors and/or other electronic devices.

BACKGROUND

[0002] For many real-world phenomena to be analyzed electronically, the phenomena may first be measured or detected in an analog form, then converted into a digital form that may be analyzed using various types of digital information processing. As one example, an image sensor may include a two-dimensional array of image sensing pixels arranged in rows and columns. Each of these pixels may include a photosensitive layer that receives incident photons (light) and converts the photons into electrical charge that may be read out row by row (with the various columns for each row being read out using parallel circuitry). To analyze and process the information captured by these pixels, charge captured by each pixel (or an analog voltage signal indicative of that charge) may be converted into a digital value using analog-to-digital converter circuitry. Image sensors utilizing such analog-to-digital converter circuitry may be included as part of various systems and devices including digital cameras, vehicles, security systems, robotic systems, mobile devices, and so forth. Analog-to-digital conversion circuitry may also be used in various other types of applications and use cases other than image sensing. For instance, these and other systems and devices may use analog-to-digital circuitry to monitor and/or analyze sound, temperature, biological vital signs, control signals, environmental conditions, and various other phenomena.

SUMMARY

[0003] Various implementations of a single-ended analog-to-digital converter input stage with transistor-based capacitance are described herein. Many conventional analog-to-digital converter circuits require a differential input signal having a different voltage range than the original single-ended analog signal that is to be converted. Accordingly, these conventional analog-to-digital converter circuits have required a pre-amplification buffer before their input stage to convert the single-ended signal into a differential signal in the desired voltage range. When implemented with discrete components, this has typically required an extra chip between the analog sensing device that generates the analog signal and the analog-to-digital converter chip performing the conversion. When implemented within an integrated circuit, this pre-amplification may be performed on-chip, but still requires power and area from potentially very stringent power and area budgets of the integrated circuit. In contrast, analog-to-digital converter circuitry described herein is configured with an input stage that has various advantages over these conventional designs. First, the input stage of various analog-to-digital converter circuits described herein may be configured to passively (without using additional power) and quickly convert a single-ended input signal into a differential signal that can be used in the analog-to-digital conversion. Second, the input stage passively and quickly shifts the input voltage level to a desired range in which the analog-to-digital conversion can be performed. Third, the input stage operates in the charge domain (such that charge, rather than voltage, is the quantity conserved and analyzed) and reduces the capacitance during the conversion procedure so as to provide a voltage boost that helps make the circuit less sensitive to noise in spite of it being low power and high speed.

[0004] To address the challenges described above and provide the advantages mentioned, systems and methods described herein implement a single-ended analog-to-digital converter input stage that uses transistor-based capacitance, such as may be provided by Complementary Metal-Oxide-Semiconductor (CMOS) transistors or other suitable transistors as may serve a particular implementation. More particularly, an input stage for an analog-to-digital converter circuit may include: 1) a first transistor of a first conductivity type, the first transistor being connected between a first node and a second node; 2) a second transistor of a second conductivity type, the second transistor being connected between the first node and a third node; and 3) a set of switches.

[0005] Transistors may be constructed with different semiconductor materials to give the transistors different properties. For example, certain transistors may be built to have a P-type conductivity while others may be built to have an N-type conductivity. If CMOS technology is used for the transistors, a transistor with P-type conductivity may be implemented by a P-channel Metal-Oxide-Semiconductor (PMOS) transistor, while a transistor with N-type conductivity may be implemented by an N-channel Metal-Oxide-Semiconductor (NMOS) transistor. Referring to the input stage described above, the conductivity types of the first and second transistors may be the same or different, such that any combination of transistors may be used in a particular implementation. For example, as will be described an illustrated in more detail below, a P-type transistor (such as a PMOS transistor) and an N-type transistor (such as an NMOS transistor) may be used in certain implementations, while, in other implementations, two transistors of the same conductivity type (two P-type transistors, two N-type transistors, etc.) may be used. In still other implementations, combinations of more than two transistors may be used in accordance with principles described and illustrated herein.

[0006] When manipulated by a controller, the switches are configured to connect: (a) the first node to either an input node or a first output node, (b) the second node to either a first reference node or a second output node, and (c) the third node to either a second reference node or the second output node. As will be described and illustrated in more detail, the manipulation of the switches may serve to put the input stage into different configurations at different times to thereby perform the desirable functionality described herein.

[0007] As mentioned above, any combination of conductivity types may be employed in the first and second transistors of an input stage such as described above. These combinations will be referred to herein by the letters P and N, as noted more specifically below. The way that the transistors are connected between the first, second, and third nodes may depend on the combination of transistors being used. As a first example (a PN combination), the first conductivity type may be a P-type and the second conductivity type may be an N-type. In this example, the first transistor may be connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node. The second transistor may then be connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node. As a second example (a PP combination), the first conductivity type and the second conductivity type may both be P-type. In this example, the first transistor may be connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node. The second transistor may then be connected between the first node and the third node by a second source and a second drain of the second transistor both connecting to the first node and a second gate of the second transistor connecting to the third node. As a third example (an NP combination), the first conductivity type may be an N-type and the second conductivity type may be a P-type. In this example, the first transistor may be connected between the first node and the second node by a first source and a first drain of the first transistor both connecting to the first node and a first gate of the first transistor connecting to the second node. The second transistor may then be connected between the first node and the third node by a second source and a second drain of the second transistor both connecting to the first node and a second gate of the second transistor connecting to the third node. As a fourth example (an NN combination), the first conductivity type and the second conductivity type may both be an N-type. In this example, the first transistor may be connected between the first node and the second node by a first source and a first drain of the first transistor both connecting to the first node and a first gate of the first transistor connecting to the second node. The second transistor may then be connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node.

[0008] An input stage for an analog-to-digital converter circuit, such as the example implementation described above, may include a variety of additional elements, features, characteristics, and so forth.

[0009] As one example, the input node may be configured to receive a single-ended signal as an input to the analog-to-digital converter circuit. The analog-to-digital converter circuit may implement a charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converter that operates in a charge domain instead of a voltage domain.

[0010] As another example, the input stage may be one of a plurality of input stages that share parallel access to a singular comparator and a singular digital-to-analog feedback circuit of the analog-to-digital converter circuit.

[0011] As another example, the analog-to-digital converter circuit may be implemented as part of an image readout circuit included within an image sensor integrated circuit. In this example, the input stage may be associated with a particular column of a plurality of columns within the image sensor integrated circuit, and the input stage may receive, at the input node, a single-ended column readout signal for the particular column.

[0012] As another example, the set of switches may be manipulated to put the analog-to-digital converter circuit in a sampling configuration in which: 1) the first node is connected to the input node and disconnected from the first output node; 2) the second node is connected to the first reference node and disconnected from the second output node; and 3) the third node is connected to the second reference node and disconnected from the second output node. The set of switches may also be manipulated to put the analog-to-digital converter circuit in a conversion configuration in which: 1) the first node is connected to the first output node and disconnected from the input node; 2) the second node is connected to the second output node and disconnected from the first reference node; and 3) the third node is connected to the second output node and disconnected from the second reference node. In some cases, the set of switches is manipulated to put the analog-to-digital converter circuit in the conversion configuration during a conversion phase. At a first time during the conversion phase, at least one of the first transistor or the second transistor may be activated such that a transistor capacitance is enabled for use in storing a sampled charge that was captured during a sampling phase prior to the first time. At a second time during the conversion phase, both the first transistor and the second transistor may be deactivated such that the transistor capacitance is disabled, and a corresponding voltage boost is applied between the first output node and the second output node as the sampled charge is conserved. Based on the sampled charge conserved within a capacitance between the first output node and the second output node when the analog-to-digital converter circuit is in the conversion configuration, a differential signal corresponding to the sampled charge may be generated on the first output node and the second output node, and this differential signal may be received by a first input and a second input of a comparator that is further included within the analog-to-digital converter circuit (after the input stage).

[0013] As another example, the first transistor may have a first bulk terminal and the second transistor may have a second bulk terminal. The set of switches may then be further configured, when manipulated by the controller, to connect: 1) the first bulk terminal to either the first reference node or a first shifted supply node, and 2) the second bulk terminal to either the second reference node or a second shifted supply node. In some cases, these features may be used to manipulate the set of switches to put the analog-to-digital converter circuit in a voltage-shift configuration. In the voltage-shift configuration: 1) the first bulk terminal may be connected to the first shifted supply node; 2) the second bulk terminal may be connected to the second shifted supply node; 3) the first node may be floating disconnected from both the input node and the first output node; 4) the second node may be floating disconnected from both the first reference node and the second output node; and 5) the third node may be floating disconnected from both the second reference node and the second output node.

[0014] In another example implementation, an analog-to-digital converter circuit includes: 1) a comparator having a first input and a second input; 2) an input stage including: (a) a first transistor of a first conductivity type, the first transistor being connected between a first node and a second node, and (b) a second transistor of a second conductivity type, the second transistor being connected between the first node and a third node; 3) a digital-to-analog feedback circuit including a series of successive-approximation capacitors; and 4) a set of switches. In this implementation, when manipulated by a controller (which may also be included in the analog-to-digital converter circuit in certain examples), the set of switches may be configured to connect: 1) the node to either an input node or the first input of the comparator, 2) the second node to either a first reference node or the second input of the comparator, 3) the third node to either a second reference node or the second input of the comparator, and 4) with either a positive polarity or a negative polarity, each capacitor of the series of successive-approximation capacitors between the first input and the second input of the comparator.

[0015] As mentioned above, the first and second transistors may each be of P-type or N-type and may be connected between the first, second, and third nodes in a manner dependent on their conductivity type. For instance, a PN combination may be used in which the first conductivity type is a P-type and the second conductivity type is an N-type. In this example, the first transistor may be connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node. The second transistor may then be connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node. In other implementations, other combinations such as PP, NP, and NN combinations of transistors may be used similarly as described above.

[0016] An analog-to-digital converter circuit, such as the example implementation described above, may include a variety of additional elements, features characteristics, and so forth.

[0017] As one example, the input node may be configured to receive a single-ended signal as an input to the analog-to-digital converter circuit. The analog-to-digital converter circuit may also implement a charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converter that operates in a charge domain instead of a voltage domain.

[0018] As another example, the set of switches may be further configured, when manipulated by the controller, to connect each capacitor of the series of successive approximation capacitors between a first precharge reference node and a second precharge reference node (such as during a phase prior to a conversion phase).

[0019] As another example, the set of switches may be manipulated to put the analog-to-digital converter circuit in a conversion configuration in which: 1) the first node is connected to the first input of the comparator and disconnected from the input node; 2) the second node is connected to the second input of the comparator and disconnected from the first reference node; 3) the third node is connected to the second input of the comparator and disconnected from the second reference node; and 4) each capacitor of the series of successive approximation capacitors is connected with either the positive polarity or the negative polarity between the first input and the second input of the comparator so as to inversely match a sampled charge captured by the input stage when the analog-to-digital converter circuit was in a sampling configuration prior to being put in the conversion configuration.

[0020] As another example, the analog-to-digital converter circuit may further include an additional input stage that is configured to share parallel access, with the input stage mentioned above, to the comparator and the digital-to-analog feedback circuit.

[0021] In yet another example implementation, a method is performed for an analog-to-digital

converter circuit that includes an input stage with a first transistor of a first conductivity type connected between a first node and a second node and a second transistor of a second conductivity type connected between the first node and a third node. The method may include operations such as: 1) manipulating a set of switches to put the analog-to-digital converter circuit in a sampling configuration, and 2) manipulating the set of switches to put the analog-to-digital converter circuit in a conversion configuration. When the analog-to-digital converter circuit is in the sampling configuration, 1) the first node is connected to an input node and disconnected from a comparator, 2) the second node is connected to a first reference node and disconnected from the comparator, and 3) the third node connected to a second reference node and disconnected from the comparator. In contrast, when the analog-to-digital converter circuit is in the conversion configuration, 1) the first node is disconnected from the input node and connected to a first input of the comparator, 2) the second node is disconnected from the first reference node and connected to a second input of the comparator, and 3) the third node is disconnected from the second reference node and connected to the second input of the comparator.
[0022] As mentioned above, the first and second transistors may each be of P-type or N-type and may be connected between the first, second, and third nodes in a manner dependent on their conductivity type. For instance, a PN combination may be used in which the first conductivity type is a P-type and the second conductivity type is an N-type. In this example, the first transistor may be connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node. The second transistor may then be connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node. In other implementations, other combinations such as PP, NP, and NN combinations of transistors may be used similarly as described above.
[0023] A method such as the example method described above may include a variety of additional elements, features characteristics, and so forth.
[0024] For example, the method may be performed with the input node being configured to receive a single-ended signal as an input to the analog-to-digital converter circuit, and with the analog-to-digital converter circuit implementing a charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converter that operates in a charge domain instead of a voltage domain.
[0025] As another example, the method may further include manipulating the set of switches to put the analog-to-digital converter circuit in a voltage-shift configuration (for instance, after the sampling configuration and before the conversion configuration). In the voltage-shift configuration: 1) a first bulk terminal of the first transistor may be connected to a first shifted supply node; 2) a second bulk terminal of the second transistor may be connected to a second shifted supply node; 3) the first node may be floating disconnected from both the input node and the first input of the comparator; 4) the second node may be floating disconnected from both the first reference node and the second input of the comparator; and 5) the third node may be floating disconnected from both the second reference node and the second input of the comparator.
[0026] As another example, the method may be performed with the analog-to-digital converter circuit being implemented as part of an image readout circuit included within an image sensor integrated circuit, such that the manipulating of the set of switches to put the analog-to-digital converter circuit in the sampling configuration and in the conversion configuration may be performed as steps of an image readout procedure performed by the image readout circuit.
[0027] The details of these and other implementations are set forth in the accompanying drawings and the description below. Other features will also be apparent from the following description, drawings, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. **1** shows a block diagram of an illustrative analog-to-digital converter circuit having a single-ended input stage in accordance with principles described herein.

[0029] FIG. **2**A shows an example schematic diagram of the input stage of the analog-to-digital converter circuit of FIG. **1** in accordance with principles described herein.

[0030] FIG. **2**B shows illustrative combinations of transistors that may be used in an input stage of an analog-to-digital converter circuit in accordance with principles described herein.

[0031] FIG. **3** shows an example schematic diagram of the digital-to-analog feedback circuit of the analog-to-digital converter circuit of FIG. **1** in accordance with principles described herein.

[0032] FIG. **4** shows an example method that may be performed in relation to the analog-to-digital converter circuit of FIG. **1** in accordance with principles described herein.

[0033] FIG. **5**A shows a schematic diagram of an example sampling configuration of a single-ended analog-to-digital converter input stage with transistor-based capacitance in accordance with principles described herein.

[0034] FIG. **5**B shows a schematic diagram of an example voltage-shift configuration of the single-ended analog-to-digital converter input stage with transistor-based capacitance in accordance with principles described herein.

[0035] FIG. **5**C shows a schematic diagram of an example conversion configuration of the single-ended analog-to-digital converter input stage with transistor-based capacitance in accordance with principles described herein.

[0036] FIG. **6** shows illustrative aspects of how an input stage and a digital-to-analog feedback circuit may interoperate during various phases of an analog-to-digital conversion procedure in accordance with principles described herein.

[0037] FIG. **7**A shows certain aspects of an illustrative image sensor integrated circuit within which implementations of analog-to-digital converter circuitry described herein may be included in accordance with principles described herein.

[0038] FIGS. **7**B and **7**C show certain aspects of an illustrative analog-to-digital converter circuit implementation that includes a plurality of input stages sharing access to a digital-to-analog feedback circuit and a comparator in accordance with principles described herein.

DETAILED DESCRIPTION

[0039] Various implementations of a single-ended analog-to-digital converter input stage with transistor-based capacitance (such as provided by Complementary Metal-Oxide-Semiconductor (CMOS) transistors or other suitable transistors) are described herein. As mentioned above, there are a variety of uses for analog-to-digital conversion circuitry in modern electronic applications. One example that will be described herein involves image sensors that create analog signals representative of detected light and convert those analog signals to digital values that can be processed digitally to generate and manipulate digital images. It will be understood, however, that many other example applications and use cases, in which analog signals are generated based on various phenomena, may be similarly configured to make use of analog-to-digital converter circuits such as those described herein.

[0040] Regardless of the application or use case, it may often be the case that an analog signal that is to be converted into a digital signal is initially generated as a single-ended signal from some type of analog sensor. For instance, a column readout signal associated with a particular pixel within an image sensor may generally be a single-ended voltage signal that is to be converted into a digital value for that pixel. This analog signal may also tend to be generated within a particular voltage range in which the sensor operates (in a range of the photosensitive light detection element of the pixel in the image sensor example). A technical problem therefore may arise when this analog signal is to be processed by an analog-to-digital converter circuit that is configured to operate with differential signals in a different voltage range (such as a lower voltage range). Conventionally, this technical problem has been addressed by adding an active amplifier stage prior to the analog-to-digital converter circuit to prepare the analog signal by shifting its voltage, converting it to a differential signal, and so forth. This type of amplifier stage may consume power and area on an integrated circuit (or may require a separate chip in a discrete component example) where such power and area resources may be scarce.

[0041] Accordingly, analog-to-digital converter circuits described herein include an input stage that uses transistor-based capacitance to passively address each of these technical challenges in a manner that eliminates the need for a pre-amplifier stage, or at least mitigates the problem by significantly reducing what pre-amplification is performed on the signal enroute to the analog-to-digital converter circuit. Specifically, as will be further illustrated and described below, analog-to-digital converter circuits described herein may implement charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converters that operate in a charge domain instead of a voltage domain. The input stage for these CS-SAR analog-to-digital converter circuits may sample the charge provided by a single-ended signal that is received as an input to the analog-to-digital converter circuit and store that sampled charge on at least two transistors that are interconnected in a particular manner based on their respective conductivity types (P-type or N-type).

[0042] As mentioned above, transistors may be constructed with different semiconductor materials to give the transistors different properties. For example, certain transistors may be built to have a P-type conductivity while others may be built to have an N-type conductivity. If CMOS technology is used for the transistors, a transistor with P-type conductivity may be implemented by a P-channel Metal-Oxide-Semiconductor (PMOS) transistor, while a transistor with N-type conductivity may be implemented by an N-channel Metal-Oxide-Semiconductor (NMOS) transistor. In some implementations of input stages described herein, the conductivity types of the first and second transistors may be the same, while in other implementations the conductivity types may be different. As enumerated above and described and illustrated in more detail below, for example, any of a PN, PP, NP, or NN combination of transistors, or other suitable combinations (such as using more than two transistors) may be employed in a given implementation. Though any of these transistor combinations may be suitable for implementing principles described herein, most examples described and illustrated herein relate to PN combinations of CMOS transistors. In these examples, a first transistor has a P-type conductivity and is implemented as a PMOS transistor, and a second transistor has an N-type conductivity transistor and is implemented as an NMOS transistor. In spite of this focus on PN combinations in the examples, it will be understood that transistor technologies other than CMOS and transistor combinations other than a PN combination (in which the first transistor is P-type and the second transistor is N-type) may additionally or alternatively employed.

[0043] By reconfiguring the connections between any suitable transistor combination described herein using a set of controller-manipulated switches, the sampled charge captured from the single-ended input signal may then be represented by a differential signal passively constructed from the capacitance of the transistors. This differential signal may then be used by the CS-SAR analog-to-digital converter circuit in its conversion phase (after the sampling is complete).

[0044] Additionally, if the voltage of the input signal is not within a desirable voltage range for the CS-SAR analog-to-digital converter circuit, the combination of transistors used as sampling capacitors for the input may also be used to perform a passive and fast voltage shift to a desirable voltage range after the sampling phase is complete and before the conversion phase begins. During conversion, as a digital-to-analog feedback circuit and a comparator work with the controller to match the sampled charge and determine a digital value corresponding to the sampled analog charge, the transistors automatically deactivate as the differential voltage on them progressively attenuates toward the midscale value. This deactivation provides yet another advantage for the circuit that a conventional capacitor would not provide. By shutting off their capacitance (while the circuit still conserves charge), the transistor-based capacitors provide a voltage boost on the differential signal that helps determine an accurate digital value and makes the circuit less sensitive to noise such as noise generated by the active comparator circuitry.

[0045] Accordingly, implementations of the single-ended input stage described herein with transistor-based capacitance allow for all the advantages of CS-SAR analog-to-digital converter circuits, including low power, high speed, and compact area. At the same time, various compromises, such as noise sensitivity as the voltage produced by the sample charge attenuates during conversion, pre-amplifier stages needed to actively shift voltage levels and/or convert single-ended signals to differential signals, etc., that might otherwise accompany the benefits of CS-SAR analog-to-digital converter circuits may be avoided or mitigated by use of this input stage.

[0046] Various implementations will now be described in more detail with reference to the figures. It will be understood that the particular implementations described below are provided as non-limiting examples and may be applied in various situations. Additionally, it will be understood that other implementations not explicitly described herein may also fall within the scope of the claims set forth below. Implementations of analog-to-digital converter circuits and associated single-ended input stages with transistor-based capacitance described herein may result in any or all of the technical benefits mentioned above, as well as various additional technical benefits that will be described and/or made apparent below.

[0047] FIG. **1** shows a block diagram of an illustrative analog-to-digital converter (ADC) circuit **100** having a single-ended input stage in accordance with principles described herein. As will be described and set forth below, analog-to-digital converter circuits such as ADC circuit **100** may be implemented in a variety of ways and with a variety of optional features that may be used in any combination as may serve a particular implementation. The block diagram of ADC circuit **100** in FIG. **1** is therefore presented as a high-level representation of certain features that may be common to many or all of the analog-to-digital converter circuit implementations described herein, and various additional implementations with different specific features and combinations of features will be described and illustrated below utilizing the same or related reference numbers as those introduced in relation to FIG. **1**. In particular, certain components illustrated in ADC circuit **100** may be illustrated in more detail in later figures, as indicated in FIG. **1** by the dashed line boxes associated with these components (boxes indicating “[FIGS. **2**A-**2**B]”, “[FIG. **3**]”, etc.).

[0048] As shown in FIG. **1**, ADC circuit **100** may include a comparator **102** having a first input **104-1** and a second input **104-2**. As will be described, these inputs may carry a differential signal that is generated by an input stage **106** that includes a first transistor of a first conductivity type that is connected between a first node and a second node, as well as a second transistor of a second conductivity type that is connected between the first node and a third node. While these transistors and nodes are not explicitly shown in FIG. **1**, the dashed box on input stage **106** indicates that more detail (including these transistors and their various gate, source, and drain terminals) is illustrated for this component in FIGS. **2**A-**2**B. As mentioned above, it will be understood that the first transistor may be of a first conductivity type, the second transistor may be of a second conductivity type (the same type or a different type as the first conductivity type), and any combination of N-type and P-type may be used if connected properly (as described and illustrated in more detail below). While examples described and illustrated herein involve CMOS transistors, where PMOS transistors are used for P-type transistors and NMOS transistors are used for N-type transistors, it will be understood that the transistors may be implemented using transistor technologies other than CMOS.

[0049] The inputs **104-1** and **104-2** to comparator **102** are also shown to be coupled with a digital-to-analog (DAC) feedback circuit **108** that may include a series of successive-approximation capacitors. Here again, while these successive approximation capacitors are not explicitly shown in

FIG. **1**, the dashed box on DAC feedback circuit **108** indicates that more detail (including this series of capacitors) is illustrated for this component in FIG. **3**. ADC circuit **100** is further shown to include a set of switches **110** that are configured to be manipulated by a controller **112**. The set of switches **110** is drawn as spanning across input stage **106** and DAC feedback circuit **108** to illustrate that one or more switches **110** included in the set that controller **112** manipulates may be included in each of these components. Specific switches within input stage **106** and DAC feedback circuit **108** will be illustrated in schematic diagrams below, such as in FIGS. **2**A and **3**.

[0050] Controller **112** is drawn as being halfway in and halfway out of ADC circuit **100** to illustrate that this component may or may not be included as part of ADC circuit **100**. For instance, in certain implementations, controller **112** may be implemented by circuitry that is part of the ADC circuit **100** and is located near these other components. In other implementations, controller **112** may be implemented separately and apart from ADC circuit **100** so as to, for example, control a plurality of similar analog-to-digital converter circuits (including ADC circuit **100**) and/or control other functions of other circuits besides the analog-to-digital converter circuits. In either type of implementation, FIG. **1** shows that controller **112** may perform manipulations of the set of switches **110** in accordance with a method **114** that will be described in more detail with reference to FIG. **4**.

[0051] ADC circuit **100** is further shown in FIG. **1** to include one or more inputs (collectively represented by the arrow labeled “Input”) such as an input node **116**, a first reference node **118-1**, and a second reference node **118-2**. As shown, input node **116** may be configured to receive a single-ended signal as an input to ADC circuit **100**, while input stage **106** may passively convert this single-ended signal into a differential signal on first input **104-1** and second input **104-2**, thereby supporting an implementation of ADC circuit **100** as a charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converter that operates in the charge domain instead of the voltage domain. Given these inputs, the set of switches **110** may be positioned within input stage **106** such that: 1) the first node may connect to either input node **116** or first input **104-1** of comparator **102**; 2) the second node may connect to either first reference node **118-1** or second input **104-2** of comparator **102**; 3) the third node may connect to either second reference node **118-2** or second input **104-2** of comparator **102**; and 4) each capacitor of the series of successive-approximation capacitors (within DAC feedback circuit **108**) may connect with either a positive polarity or a negative polarity between first input **104-1** and second input **104-2** of comparator **102**. Various configurations (in which the set of switches **110** is manipulated in different ways) will be described and illustrated below to show the various possibilities this architecture provides.

[0052] ADC circuit **100** is further shown in FIG. **1** to include one or more outputs (collectively represented by the arrow labeled “Output”). In particular, a digital output **120** is shown to emerge from controller **112**. Digital output **120** may provide, using any suitable number of bits, a digital representation of the single-ended input signal sampled on input node **116** after this analog input is converted to a digital value. Various elements shown in FIG. **1** to be included within or otherwise associated with ADC circuit **100** will now be described in relation to FIGS. **2**A-**4**.

[0053] FIG. **2**A shows an example schematic diagram of input stage **106** of ADC circuit **100**. As shown, input stage **106** includes a first transistor **202-1** with a first conductivity type and a second transistor **202-2** with a second conductivity type. As shown, transistor **202-1** is connected between a first node **206-1** and a second node **206-2**, while transistor **202-2** is connected between first node **206-1** and a third node **206-3**. As described and illustrated in more detail below, terminals of these transistors **202** may be connected to the nodes **206** in a manner that depends on their conductivity type. It is also noted that each transistor is this example is shown to include a bulk terminal (also referred to as a body terminal and labeled ‘B**1**’ and ‘B**2**’ in FIG. **2**A) that may also be connected to different reference voltages based on the objectives of a particular implementation and based on the respective conductivity types of the transistors. For example, objectives may include a desired voltage shift including one or more voltages, among others. As will be described in more detail below, it will be understood that the manner in which the bulk terminals are shown to be connected in the various examples illustrated herein (including the example of FIG. **2**A) correspond to certain assumptions about the desired voltage shift and the transistor combination being employed in these examples. As will be set forth in more detail below, implementations involving other transistor combinations and/or desired values of voltage shift may also employ different connections for the bulk terminals than those shown in these figures (e.g., connections allow the bulk terminals to optionally connect to yet another reference voltage option, etc.).

[0054] Input stage **106** is also shown to include various switches of the set of switches **110** described above. Each of these is labeled with a unique number **110**-*x*, where 'x' represents different identifier numbers to differentiate the various switches. As shown, input stage **106** includes input node **116** to receive the single-ended input, as well as a first output node **204-1** and a second output node **204-2** that will carry the differential signal to comparator **102**. As such, it is noted in FIG. **2**A that the first and second output nodes **204-1** and **204-2** of input stage **106** correspond to (or are equivalent to) the nodes labeled in FIG. **1** as first and second inputs **104-1** and **104-2** of comparator **102**.

[0055] As mentioned above, when manipulated by a controller such as controller **112**, switches **110** may allow for various terminals of transistor **202-1** and transistor **202-2** (also referred to collectively as transistors **202**) to be connected in a variety of different configurations. FIG. **2**A illustrates this in more detail. Specifically, as shown, a switch **110-1** and a switch **110-2** may allow first node **206-1** to connect to either input node **116** (when switch **110-1** is closed and switch **110-2** is open) or first output node **204-1** (when switch **110-1** is open and switch **110-2** is closed). A switch **110-3** and a switch **110-4** may allow second node **206-2** to connect to either first reference node **118-1** (when switch **110-3** is closed and switch **110-4** is open) or second output node **204-2** (when switch **110-3** is open and switch **110-4** is closed). Similarly, a switch **110-5** and a switch **110-6** may allow third node **206-3** to connect to either second reference node **118-2** (when switch **110-5** is closed and switch **110-6** is open) or second output node **204-2** (when switch **110-5** is open and switch **110-6** is closed).

[0056] Voltages provided at first reference node **118-1** and second reference node **118-2** may be any suitable reference voltages as may serve a particular implementation. In some implementations, these reference voltages may be wider than the voltage range at which the input signal is provided (on input node **116**) by at least a threshold voltage so as to ensure that, when the input signal is sampled, both transistors **202** are activated. For example, if the input voltage range expected on input node **116** is 1.0V-2.2V and transistors **202** each have a threshold voltage of 0.3V, first reference node **118-1** may provide a reference voltage of at least 2.5V and second reference node **118-2** may provide a reference voltage of no more than 0.7V.

[0057] As has been described, any transistor combination of two or more transistors **202** (transistors **202-1** and **202-2** in FIG. **2**A) may be used in a given implementation of input stage **106**. To illustrate, a dashed box around transistors **202-1** and **202-2** (as well as at least parts of nodes **206-1**, **206-2**, and **206-3**) is drawn to represent a particular transistor combination **210** selected from a variety of potential combinations that will now be illustrated and described with reference to FIG. **2**B, which shows illustrative transistor combinations that may be used in various implementations of input stage **106** in accordance with principles described herein.

[0058] As shown in FIG. **2**B, four transistor combinations **210** are labeled using letters 'P' and/or 'N' to indicate which conductivity type is used for transistor **202-1** and for transistor **202-2**. As mentioned above, a similar notation with 'P's and/or 'N's has been used to refer to, for example, PN combinations, PP combinations, and so forth. Each of these transistor combinations **210**-PN (a PN combination), **210**-PP (a PP combination), **210**-NN (a NN combination), and NP (an NP combination) are illustrated in FIG. **2**B and will now be described.

[0059] In transistor combination **210**-PN, the conductivity type of transistor **202-1** is a P-type and the conductivity type of transistor **202-2** is an N-type. As such, these transistors are drawn, respectively, as a PMOS transistor (labeled as transistor **202-1**P) and as an NMOS transistor (labeled as transistor **202-2**N). As shown in the example of transistor combination **210-**PN, first transistor **202-1**P is connected between first node **206-1** and second node **206-2** by a first gate (labeled 'G**1**') of first transistor **202-1**P connecting to first node **206-1** and a first source (labeled 'S**1**') and a first drain (labeled 'D**1**') of first transistor **202-1**P both connecting to second node **206-2**. A bulk terminal of first transistor **202-1**P is also shown (labeled 'B**1**') and will be described in more detail below. Second transistor **202-2**N is shown to be connected between first node **206-1** and third node **206-3** by a second gate (labeled 'G**2**') of second transistor **202-2**N connecting to first node **206-1** and a second source (labeled 'S**2**') and a second drain (labeled 'D**2**') of second transistor **202-2**N both connecting to third node **206-3**. A bulk terminal of second transistor **202-2**N is also shown (labeled 'B**2**') and will be described in more detail below.

[0060] In transistor combination **210-**PP, the conductivity type of both transistors **202-1** and **202-2** is a P-type. As such, these transistors are drawn, respectively, as PMOS transistors (labeled as transistors **202-1**P and **202-2**P). As shown in the example of transistor combination **210-**PP, first transistor **202-1**P is connected between first node **206-1** and second node **206-2** by a first gate (labeled 'G**1**') of first transistor **202-1**P connecting to first node **206-1** and a first source (labeled 'S**1**') and a first drain (labeled 'D**1**') of first transistor **202-1**P both connecting to second node **206-2**. A bulk terminal of first transistor **202-1**P is again shown (labeled 'B**1**') and will be described in more detail below. Second transistor **202-2**P is shown to be connected between first node **206-1** and third node **206-3** by a second source (labeled 'S**2**') and a second drain (labeled 'D**2**') of second transistor **202-2**P both connecting to first node **206-1** and a second gate (labeled 'G**2**') of second transistor **202-2**P connecting to third node **206-3**. A bulk terminal of second transistor **202-2**P is also shown (labeled 'B**2**') and will be described in more detail below.

[0061] In transistor combination **210-**NN, the conductivity type of both transistors **202-1** and **202-2** is an N-type. As such, these transistors are drawn, respectively, as NMOS transistors (labeled as transistors **202-1**N and **202-2**N). As shown in the example of transistor combination **210-**NN, first transistor **202-1**N is connected between first node **206-1** and second node **206-2** by a first source (labeled 'S**1**') and a first drain (labeled 'D**1**') of first transistor **202-1**N both connecting to first node **206-1** and a first gate (labeled 'G**1**') of first transistor **202-1**N connecting to second node **206-2**. A bulk terminal of first transistor **202-1**N is again shown (labeled 'B**1**') and will be described in more detail below. Second transistor **202-2**N is shown to be connected between first node **206-1** and third node **206-3** by a second gate (labeled 'G**2**') of second transistor **202-2**N connecting to first node **206-1** and a second source (labeled 'S**2**') and a second drain (labeled 'D**2**') of second transistor **202-2**N both connecting to third node **206-3**. A bulk terminal of second transistor **202-2**N is also shown (labeled 'B**2**') and will be described in more detail below.

[0062] In transistor combination **210-**NP, the conductivity type of transistor **202-1** is an N-type and the conductivity type of transistor **202-2** is a P-type. As such, these transistors are drawn, respectively, as an NMOS transistor (labeled as transistor **202-1**N) and as a PMOS transistor (labeled as transistor **202-2**P). As shown in the example of transistor combination **210-**NP, first transistor **202-1**N is connected between first node **206-1** and second node **206-2** by a first source (labeled 'S**1**') and a first drain (labeled 'D**1**') of first transistor **202-1**N both connecting to first node **206-1** and a first gate (labeled 'G**1**') of first transistor **202-1**N connecting to second node **206-2**. A bulk terminal of first transistor **202-1**N is again shown (labeled 'B**1**') and will be described in more detail below. Second transistor **202-2**P is shown to be connected between first node **206-1** and third node **206-3** by a second source (labeled 'S**2**') and a second drain (labeled 'D**2**') of second transistor **202-2**P both connecting to first node **206-1** and a second gate (labeled 'G**2**') of second transistor **202-2**P connecting to third node **206-3**. A bulk terminal of second transistor **202-2**P is also shown (labeled 'B**2**') and will be described in more detail below.

[0063] Returning to FIG. **2**A, along with the respective gate, source, and drain terminals described above, transistors **202** are also each shown to include the respective bulk terminals (also referred to as body terminals) that were mentioned above and that may also be reconfigurable based on certain switches of the set of switches **110**. Specifically, as shown, first transistor **202-1** may have a first bulk terminal (labeled 'B**1**') and second transistor **202-2** may have a second bulk terminal (labeled 'B**2**'). The set of switches **110** is further configured, when manipulated by controller **112**, to allow these bulk terminals to be connected in different configurations, though it will be understood that for sake of clarity and conciseness, certain configurations that could be desirable for certain circumstances (e.g., for transistor combinations other than the PN combination **210**-PN in FIG. **2**B, etc.) may not be achievable by the particular set of switches and connections illustrated in FIG. **2**A and other figures below. For example, assuming a PN transistor combination such as illustrated by transistor combination **210-**PN, FIG. **2**A shows a switch **110-7** and a switch **110-8** that allow the first bulk terminal B**1** to connect to either first reference node **118-1** (when switch **110-7** is closed and switch **110-8** is open) or a first shifted supply node **208-1** (when switch **110-7** is open and switch **110-8** is closed). Similarly, a switch **110-9** and a switch **110-10** may allow the second bulk terminal B**2** to connect to either second reference node **118-2** (when switch **110-9** is closed and switch **110-10** is open) or a second shifted supply node **208-2** (when switch **110-9** is open and switch **110-10** is closed). Other connections (e.g., such as to other supply nodes rather than to reference nodes **118-1** and **118-2**) through other switches not explicitly shown in FIG. **2**A may be used to achieve an analogous voltage shift for transistor combinations other than the PN combination assumed for this example. For instance, if transistor **202-1** were to have N-type rather than P-type conductivity, a significantly lower bulk voltage than that provided at reference node **118-1** may be desirable. Accordingly, rather than switch **110-7** connecting the bulk terminal to reference node **118-1**, this switch may optionally connect the bulk terminal to a lower voltage rail not explicitly shown in FIG. **2**.

[0064] As will be described in more detail below, the bulk terminals and their associated switches **110** may be used to implement a voltage shift for signals on both output nodes **204-1** and **204-2** in implementations where such a shift is desirable for downstream circuitry (e.g., comparator **102**, DAC feedback circuit **108**, etc.). As such, voltages provided at first shifted supply node **208-1** and second shifted supply node **208-2**, like those provided at first reference node **118-1** and second reference node **118-2**, may be any suitable reference voltages as may serve a particular implementation. For example, if it is assumed that a PN transistor combination is used for transistors **202** and that downstream circuitry is configured to operate with a supply voltage (VDD) of 1.5V and a ground voltage of 0V, first shifted supply node **208-1** could be connected to VDD (1.5V), while second shifted supply node **208-2** could be connected to ground (0V). It will be understood that these values may be different for other transistor combinations and situations.

[0065] FIG. **3** shows an example schematic diagram of DAC feedback circuit **108** of ADC circuit **100**, along with comparator **102**. As mentioned above, DAC feedback circuit **108** may include a plurality of capacitors referred to herein as a series of successive-approximation capacitors. To illustrate this, FIG. **3** shows one successive-approximation capacitor **302-**A included, along with various switches from the set of switches **110**, within a first successive-approximation-register (SAR) stage **304-**A. While not all explicitly shown, FIG. **3** illustrates that several additional SAR stages **304-**B, **304-**C, and others (as indicated by an ellipsis) similar to SAR stage **304-**A may also be included. It will be understood that each of these SAR stages **304** may include its own successive-approximation capacitor and its own switches to perform a similar function as will be described for SAR stage **304-**A. As such, the 'A' in the designators for the components shown in SAR stage **304-**A is meant to indicate that these components may have similar counterparts (which, if shown, could be referred to with designators 'B', 'C', etc.) in each of the other SAR stages **304**. Additionally, while space constraints in the figure do not allow for each switch **110** to be explicitly numbered as switch **110-***x*, the designator placed next to each switch ('**11**A', '**12**A', etc.) will be understood to differentiate that switch **110** from others and the switches will be referred to using their full designators (switches **110-11**A, **110-12**A, etc.).

[0066] DAC feedback circuit **108** is shown to be connected to inputs **104-1** and **104-2** of comparator **102**, which, as described and illustrated in FIG. **2**A (and as noted in FIG. **3**), may be the same nodes as output nodes **204-1** and **204-2** of input stage **106**. As will be described in more detail below, DAC feedback circuit **108** may be configured to precharge the series of successive-approximation capacitors **302** (i.e., successive-approximation capacitor **302-**A and other successive-approximation capacitor included in the other SAR stages **304**) while the input stage is in a sampling configuration or otherwise is not yet connected to DAC feedback circuit **108** and comparator **102**. DAC feedback circuit **108** may then be configured (as directed by controller **112**) to step through and connect each of the successive-approximation capacitors **302**, with either a positive polarity or a negative polarity, between inputs **104-1** and **104-2** of comparator **102** during a conversion phase.

[0067] Accordingly, FIG. **3** shows how the set of switches **110** may be configured (when manipulated by controller **112**) to allow each capacitor of the series of successive-approximation capacitors **302** to connect between a first precharge reference node **306-1** and a second precharge reference node **306-2**. Specifically, as shown, when a switch **110-11**A is closed (and when a switch **110-12**A and a switch **110-13**A are open), a first terminal of successive-approximation capacitor **302-**A is connected to first precharge reference node **306-1**, while when a switch **110-14**A is closed (and when a switch **110-15**A and a switch **110-16**A are open), the other terminal of successive-approximation capacitor **302-**A is connected to second precharge reference node **306-2**. The reference voltage provided at precharge reference nodes **306-1** and **306-2** may be any suitable reference voltage. For example, the precharge reference voltage may be the same VDD and ground voltage mentioned above in connection with shifted supply nodes **208-1** and **208-2**, or may be a version of that voltage (an isolated or cleaner version of the supply and ground voltages). As another example, the precharge reference voltage may be different from the VDD and ground supply voltages, such as a slightly wider voltage, a slightly narrower voltage, or the like, as may serve a particular implementation.

[0068] Once the successive-approximation capacitors are precharged (such as during a sampling phase and/or a voltage-shift phase) and a conversion phase is entered, controller **112** may manipulate switches **110** to successively connect and align each of the successive-approximation capacitors **302** between inputs **104-1** and **104-2** with either a positive polarity or a negative polarity so as to try to match the sampled charge (represented by the differential voltage put on inputs **104** by input stage **106**) more and more closely. For example, as shown, successive-approximation capacitor **302-**A may be connected between inputs **104-1** and **104-2** with a positive polarity when switches **110-12**A and **110-15**A are closed (and switches **110-11**A, **110-13**A, **110-14**A, and **110-16**A are open). Successive-approximation capacitor **302-**A may be connected between inputs **104-1** and **104-2** with a negative polarity when switches **110-13**A and **110-16**A are closed (and switches **110-11**A, **110-12**A, **110-14**A, and **110-15**A are open).

[0069] Successive-approximation capacitors **302-**A may be sized with different capacitances such that, when connected and aligned with respective polarities that bring the differential voltage on inputs **104-1** and **104-2** to a midrange value (which may occur when the charge held in the successive-approximation capacitors **302** approximately matched with the sampled charge in the transistor capacitance of transistors **202-1** and **202-2**), the respective polarities of the various capacitors collectively create a digital value corresponding to the sampled charge. For example, each successive-approximation capacitor **302** in the series may be configured with half the capacitance of the capacitor preceding it and the polarity or alignment needed to approximately match the sampled charge may represent one significant digit of the digital output.

[0070] For an N-bit CS-SAR analog-to-digital converter circuit, DAC feedback circuit **108** may include N successive-approximation capacitors **302**. As one example, a 10-bit CS-SAR analog-to-digital converter circuit may include ten capacitors. The N capacitors within DAC feedback circuit **108** may all be pre-charged to a known precharge reference voltage level (the reference voltage between precharge reference nodes **306-1** and **306-2**) prior to the conversion phase, and selectively coupled to the inputs **104-1** and **104-2** in different polarities according to the amount of charge that was sampled by input stage **106**.

[0071] In certain CS-SAR analog-to-digital converter circuits, the successive-approximation capacitors may be binary scaled so as to have a radix-2 sizing scheme. For example, an N-bit CS-SAR analog-to-digital converter circuit may include N successive-approximation capacitors with capacitance values C, 2C, 4C, 8C, 16C, . . . , 2.sup.N−1C. In other CS-SAR analog-to-digital converter circuits, the successive-approximation capacitors may instead by scaled using a sub-radix-2 sizing scheme in which, for any capacitor N with capacitance C.sub.N, its capacitance C.sub.N-1 is less than (rather than equal to) the sum of the capacitances of all smaller capacitors C.sub.N−2, C.sub.N−3, C.sub.N−4, . . . , C.sub.0. In other words, instead of a scaling factor of 2, a sub-radix-2 sizing scheme may use a scaling factor such as 1.9, 1.8, 1.7, 1.5-1.9999, or any other suitable scaling factor less than 2. For instance, using a scaling factor of 1.9, an N-bit CS-SAR analog-to-digital converter circuit may include N successive-approximation capacitors with capacitance values C, 1.90C, 3.61C (1.9.sup.2C.), 6.86C (1.9.sup.3C), 13.0C (1.9.sup.4C.), and so forth.

[0072] FIG. **4** shows an example implementation of method **114** described above in relation to FIG. **1** as being performed by controller **112** for ADC circuit **100**. While FIG. **4** shows illustrative operations **402-406** according to one implementation, it will be understood that other implementations of method **114** may omit, add to, reorder, and/or modify any of the operations **402-406** shown in FIG. **4**. In some examples, multiple operations shown in FIG. **4** or described in relation to FIG. **4** may be performed concurrently (e.g., in parallel) with one another, rather than being performed sequentially as illustrated and/or described. Each of operations **402-406** will now be described in more detail as they may be performed by an implementation of ADC circuit **100** or, more particularly, by an implementation of controller **112** that may be integrated with or implemented separately from ADC circuit **100**. Operations **402-406** each include a dashed box referencing one of FIGS. **5**A-5C, which will also be used to illustrate these operations. Alongside method **114**, an analog-to-digital conversion procedure **408** associated with the method is shown to include several phases placed along a timeline going from the top of the page to the bottom (“Time”). In the following description, phases **410-1** through **410-3** will also be referred to in connection with their respective operations **402-406**.

[0073] At operation **402**, a controller may manipulate a set of switches to put an analog-to-digital converter circuit in a sampling configuration. As illustrated above, the analog-to-digital converter circuit described in this example will be assumed to include an input stage with a first transistor of a first conductivity type connected between a first node (first node **206-1**) and a second node (second node **206-2**) and a second transistor of a second conductivity type connected between the first node and a third node (third node **206-3**). As shown, this sampling configuration may be associated with a sampling phase **410-1** that may be the first phase performed in time (prior to the other phases **410**) in analog-to-digital conversion procedure **408**. As mentioned above and as will be illustrated and described in connection with FIG. **5**A, the set of switches may be manipulated for the sampling configuration such that: 1) the first node is connected to an input node and disconnected from a comparator, 2) the second node is connected to a first reference node and disconnected from the comparator, and 3) the third node is connected to a second reference node and disconnected from the comparator.

[0074] To illustrate, FIG. **5**A shows a schematic diagram of input stage **106** in a sampling configuration **106-5**A. The same elements of input stage **106** are shown to be included in FIG. **5**A as have been described above in relation to FIGS. **2**A-**2**B. In particular, the elements of FIG. **2**A are shown to be used with a PN transistor combination implemented with a first transistor **202-1**P and a second transistor **202-2**N (connected in the manner of transistor combination **210**-PN in FIG. **2**B). However, to illustrate the sampling configuration, certain switches **110** are now shown to be closed (while others remain open). Specifically, as shown, first node **206-1** (connecting gate G**1** of transistor **202-1**P and gate G**2** of transistor **202-2**N) is shown to be connected to input node **116** and to be disconnected from the comparator (disconnected from output node **204-1** going to comparator **102**) by switch **110-1** being closed and by switch **110-2** being open. Second node **206-2** (connecting source S**1** and drain D**1** of transistor **202-1**P) is shown to be connected to reference node **118-1** and to be disconnected from the comparator (disconnected from output node **204-2** going to comparator **102**) by switch **110-3** being closed and by switch **110-4** being open. Third node **206-3** (connecting source S**2** and drain D**2** of transistor **202-2**N) is shown to be connected to reference node **118-2** and to be disconnected from the comparator (disconnected from output node **204-2** going to comparator **102**) by switch **110-5** being closed and by switch **110-6** being open. In this example, sampling configuration **106-5**A further shows that bulk terminal B**1** of transistor **202-1**P may be connected to reference node **118-1** and disconnected from shifted supply node **208-1** by switch **110-7** being closed and by switch **110-8** being open. Bulk terminal B**2** of transistor **202-2**N is similarly connected to reference node **118-2** and disconnected from shifted supply node **208-2** by switch **110-9** being closed and by switch **110-10** being open.

[0075] In this sampling configuration, charge associated with a single-ended input signal received on input node **116** may be captured and stored in the capacitance of transistors **202**. For example, the voltage between second node **206-2** (the same as reference node **118-1** in this configuration) and first node **206-1** may be greater than a threshold voltage of transistor **202-1**P, such that transistor **202-1**P may be activated and used as a capacitor (holding a charge between gate G**1** on one side and source S**1** and drain D**1** on the other side). Similarly, and in a complementary manner characteristic of CMOS circuitry, the voltage between third node **206-3** (the same as reference node **118-2** in this configuration) and first node **206-1** may be greater than a threshold voltage of transistor **202-2**N, such that transistor **202-2**N may also be activated and used as another capacitor (holding a charge between gate G**2** on one side and source S**2** and drain D**2** on the other side). Once the capacitance on these transistors **202** is charged by the single-ended input signal, sampling phase **410-1** may be complete and the sampled charge stored within this transistor-based capacitance will thereafter be conserved for use in the CS-SAR analog-to-digital converter circuit.

[0076] Returning to FIG. **4**, at operation **404**, the controller may manipulate the set of switches to put the analog-to-digital converter circuit in a voltage-shift configuration. As shown, this voltage-shift configuration may be associated with a voltage-shift phase **410-2**. In some implementations, such as is shown in the example of FIG. **4**, voltage-shift phase **410-2** may follow sampling phase **410-1** in time during analog-to-digital conversion procedure **408**. In other implementations, it will be understood that the voltage-shift phase may not be a separate phase in time, but, rather, the reconnection of the transistor bulks may be performed at the same time (or near in time) to when connections are made for a conversion configuration described in more detail below. In other words, while shown as a separate phase in FIG. **4**, it will be understood that voltage-shift phase **410-2** may be merged with conversion phase **410-3** in certain implementations. As mentioned above and as will be illustrated and described in connection with FIG. **5**B, the set of switches may be manipulated for the voltage-shift configuration such that: 1) a first bulk terminal of the first transistor is connected to a first shifted supply node; 2) a second bulk terminal of the second transistor is connected to a second shifted supply node; 3) a first node to which both transistors are connected is floating disconnected from both the input node and the first input of the comparator; 4) a second node to which the first transistor is connected is floating disconnected from both the first reference node and the second input of the comparator; and 5) a third node to which the second transistor is connected is floating disconnected from both the second reference node and the second input of the comparator.

[0077] To illustrate, FIG. **5**B shows a schematic diagram of input stage **106** in a voltage-shift configuration **106-5**B. The same elements of input stage **106** are shown to be included in FIG. **5**B as have been described and illustrated in other examples above. However, different switches **110** are now shown to be open and closed to illustrate the voltage-shift configuration. Specifically, as

shown, bulk terminal B**1** of first transistor **202-1**P is shown to be connected to first shifted supply node **208-1** and to be disconnected from reference node **118-1** by switch **110-8** being closed and by switch **110-7** being open. Bulk terminal B**2** of second transistor **202-2**N is similarly shown to be connected to second shifted supply node **208-2** and to be disconnected from reference node **118-2** by switch **110-10** being closed and by switch **110-9** being open. Other terminals besides the bulk terminals are then shown in voltage-shift configuration **106-5**B to be floating (disconnected from other parts of the circuit other than other terminals of the transistors as shown). Specifically, FIG. **5**B shows that first node **206-1** is floating disconnected from both input node **116** and output node **204-1** (leading to input **104-1** of the comparator) by switches **110-1** and **110-2** both being open. Second node **206-2** is similarly shown to be floating disconnected from both first reference node **118-1** and second output node **204-2** (leading to input **104-2** of the comparator) by switches **110-3** and **110-4** both being open. Third node **206-3** is also shown to be floating disconnected from both second reference node **118-2** and second output node **204-2** (leading to input **104-2** of the comparator) by switches **110-5** and **110-6** both being open.
[0078] In this voltage-shift configuration, the voltage associated with the charge sampled and stored by transistors **202** during the sampling phase may be quickly and passively shifted as the terminals of these transistors (other than the bulk terminals) are floating. As indicated by dashed lines outlining operation **404** and voltage-shift phase **410-2** in FIG. **4**, this operation/phase may be understood to be optional and used only in implementations where such a voltage shift is desirable. For example, as mentioned above, an image sensor application may typically use a different voltage range where pixels are sampling light and where digital data is being converted and read out. In one particular implementation, for instance, it may be desirable to shift an input voltage from being in a voltage domain with a range between 0.7V-2.5V to be processed in a voltage domain with a range between 0.0V-1.2V.
[0079] To describe how this type of passive, downward voltage shift may be achieved, it will be understood that transistors **202** do not only have the gate capacitance described above as being between the gate on one side and the source and drain on the other side (also referred to herein as a MOS capacitance). While the gate, source, and drain terminals are all floating in this configuration, transistors **202** may each further have a bulk capacitance that is not floating but, rather is connected to the respective shifted supply nodes **208-1** and **208-2**. In voltage-shift phase **410-2**, this bulk capacitance may be used to shift the voltages on the otherwise floating transistors to the voltage range framed by the voltages on shifted supply nodes **208-1** and **208-2**. Specifically, due to capacitive coupling with the bulk capacitance, the voltages on these floating nodes will follow the bulk or body of transistors **202** to be centered within the desired voltage range. Alternatively, if no voltage shift is desired for a given implementation, this phase may be dispensed with, and the bulk terminals may be unconnected or may remain connected to respective reference nodes **118-1** and **118-2** (eliminating the need for switches **110-7** through **110-10**).
[0080] It will also be understood that a similar or the same effect as described above in relation to the voltage-shift configuration may be achieved by connecting the bulk terminals in the manner described while then immediately connecting the first node to the first output node (rather than letting it float disconnected from any other node) and connecting the second and third nodes to the second output node (rather than letting them float disconnected from any other node) in the manner described herein for the conversion configuration (see, e.g., the description of FIG. **5**C below). In other words, if a voltage shift is desired, this shift may be achieved by properly connecting the bulk terminals and then by either allowing the transistors to float as the voltage shifts or immediately connecting them in the conversion configuration without an explicit period of time where the transistors float disconnected.
[0081] Returning to FIG. **4**, at operation **406**, the controller may manipulate the set of switches to put the analog-to-digital converter circuit in a conversion configuration. As shown, this conversion configuration may be associated with a conversion phase **410-3** that may follow sampling phase

410-1 and voltage-shift phase **410-2** in time during analog-to-digital conversion procedure **408**. In some examples, as mentioned above, voltage-shift phase **410-2** may be implemented concurrently (simultaneously, partially overlapping, etc.) with conversion phase **410-3**. As mentioned above and as will be illustrated and described in connection with FIG. **5**C, the set of switches may be manipulated for the conversion configuration such that: 1) the first node is disconnected from the input node and connected to a first input of the comparator, 2) the second node is disconnected from the first reference node and connected to a second input of the comparator, and 3) the third node is disconnected from the second reference node and connected to the second input of the comparator.

[0082] To illustrate, FIG. **5**C shows a schematic diagram of input stage **106** in a conversion configuration **106-5**C. The same elements of input stage **106** are shown to be included in FIG. **5**C as have been described and illustrated in other examples above. However, different switches **110** are now shown to be open and closed to illustrate the conversion configuration. Specifically, as shown, first node **206-1** (connecting gate G**1** of transistor **202-1**P and gate G**2** of transistor **202-2**N) is shown to be connected to first output node **204-1** (to go to input **104-1** of the comparator) and to be disconnected from input node **116** by switch **110-1** now being open while switch **110-2** is closed. Second node **206-2** (connecting source S**1** and drain D**1** of transistor **202-1**P) is shown to be connected at source-drain node **206**-SD**1**) to output node **204-2** (to go to input **104-2** of the comparator) and to be disconnected from reference node **118-1** by switch **110-3** now being open while switch **110-4** is closed. Third node **206-3** (connecting source S**2** and drain D**2** of transistor **202-2**N) is also shown to be connected to output node **204-2** (to go to input **104-2** of the comparator) and to be disconnected from reference node **118-2** by switch **110-5** now being open while switch **110-6** is closed. In this example, sampling configuration **106-5**C further shows that bulk terminal B**1** of transistor **202-1**P continues to be connected to shifted supply node **208-1** and disconnected from reference node **118-1** (as was the case in voltage-shift configuration **106-5**B, assuming an example implementation that uses the optional voltage-shift phase) by switch **110-7** being open and by switch **110-8** being closed. Likewise, bulk terminal B**2** of transistor **202-2**N is shown to still be connected to shifted supply node **208-2** and disconnected from reference node **118-2** by switch **110-9** being open and by switch **110-10** being closed.

[0083] In this conversion configuration, the voltage associated with the charge that was sampled and stored by transistors **202** during the sampling phase (and that may have been shifted during the voltage-shift phase) is now connected to output nodes **204-1** and **204-2** so as to be presented at inputs **104-1** and **104-2** of comparator **102**. More particularly, FIG. **5**C shows that in the conversion configuration, the set of switches **110** may be manipulated such that: 1) the first gate and the second gate are both connected to first input **104-1** of the comparator (and disconnected from the input node); 2) the first source and the first drain are both connected to second input **104-2** of the comparator (and disconnected from the first reference node); and 3) the second source and the second drain are both connected to second input **104-2** of the comparator (and disconnected from the second reference node).

[0084] By the time the switches are manipulated in this way, it is noted the input charge has been sampled and is stored in the MOS capacitance of transistor **202-1**P and/or transistor **202-2**N (or whatever transistor combination is used in a particular implementation). Accordingly, when second node **206-2** and third node **206-3** are suddenly connected at second output node **204-2** (so as to both feed into comparator input **104-2**), the stored charge is conserved and, due to the capacitance between output node **204-1** (now connected to first node **206-1**) on one side and output node **204-2** (now connected to second node **206-2** and third node **206-3**) on the other side, a differential signal is generated between output nodes **204-1** and **204-2**. As will now be illustrated and described in more detail in relation to FIG. **6**, this differential signal arises automatically during the conversion phase based on the charge sampled during the sampling phase. More particularly, based on a sampled charge conserved within a capacitance between output nodes **204-1** and **204-2** when ADC circuit **100** is in the conversion configuration (the MOS capacitance of transistors **202**), a differential signal corresponding to the sampled charge is generated on output nodes **204-1** and **204-2** and this differential signal is received by inputs **104-1** and **104-2** of comparator **102**. As described above, certain analog-to-digital converter circuits, including CS-SAR analog-to-digital converter circuits described herein, may function based on a differential signal. Accordingly, this fast and passive way of generating a differential input signal without an additional differential conversion stage or pre-amplification circuit is highly efficient and may provide various advantages (relating to power usage, area, speed, etc.) as have been described.

[0085] While FIGS. **5**A-**5**C illustrated specifically how some certain switches of the set of switches **110** may be manipulated during the different phases of analog-to-digital conversion procedure **408**, it has been shown that additional switches **110**, which may also be manipulated by controller **112** in different ways throughout analog-to-digital conversion procedure **408**, are also implemented within DAC feedback circuit **108** (see, for example, switches **110-11**A through **110-16**A in FIG. **3**). It will be understood that these additional switches may be manipulated such that successive-approximation capacitors **302** may be precharged prior to the conversion phase **410-3** (e.g., during sampling phase **410-1** and/or voltage-shift phase **410-2**), and such that the capacitors **302** may successively be added between the comparator inputs during the course of conversion phase **410-3**. By the end of conversion phase **410-3**, each capacitor **302** of the series of successive approximation capacitors may be connected with either the positive polarity or the negative polarity between first input **104-1** and second input **104-2** of comparator **102** so as to inversely match the sampled charge captured by input stage **106** when ADC circuit **100** was in the sampling configuration prior to being put in the conversion configuration.

[0086] To illustrate, FIG. **6** shows certain aspects of how input stage **106** and DAC feedback circuit **108** may interoperate during various phases of the analog-to-digital conversion procedure **408** described above. Certain principles now described arise from ADC circuit **100** implementing a CS-SAR analog-to-digital converter that operates in a charge domain instead of a voltage domain. Other aspects arise from the unique use of CMOS-based capacitance in the single-ended input stage described and illustrated above.

[0087] As shown in FIG. **6**, a MOS capacitance **602** provided by the CMOS transistors in input stage **106** may capture and store a sampled charge **604** between the inputs **104-1** and **104-2** of comparator **102**. This sampling of the input to capture sampled charge **604** may be performed during sampling phase **410-1**, as has been described. A graph **606** including a set of waveforms to show the voltage on various nodes during the phases **410** of procedure **408** illustrates this sampling phase **410-1** near the left of the graph. Specifically, a waveform **608-1** will be understood to correspond to the voltage (on the y-axis), with respect to time (on the x-axis), during phases **410** on first node **206-1**. Similarly, a waveform **608-2** will be understood to correspond to the voltage on second node **206-2**, while a waveform **608-3** will be understood to correspond to the voltage on third node **206-3**.

[0088] During sampling phase **410-1**, graph **606** shows that waveform **608-2** (attached to the source and drain of first transistor **202-1**P and to reference node **118-1**, as shown in FIG. **5**A) has a voltage higher than the voltage of waveform **608-1** (attached to the gates of both transistors **202** and to input node **116**, as further shown in FIG. **5**A), which itself is higher than the voltage of waveform **608-3** (attached to the source and drain of second transistor **202-2**N and to reference node **118-2**, as further shown in FIG. **5**A). As has been described, the single-ended input signal coming in on input node **116** may charge up MOS capacitance **602** during this sampling phase **410-1** such that whatever sampled charge **604** is captured during this phase will be conserved thereafter.

[0089] During voltage-shift phase **410-2**, graph **606** shows that all three waveforms **608-1**, **608-2**, and **608-3** shift downward in voltage as the source, drain, and gate terminals of both transistors **202** are made to float while the bulk terminals are connected to the shifted supply nodes **208-1** and **208-2** (as was illustrated and described in relation to FIG. **5**B). The waveforms still have a similar relationship to one another during this phase, but their voltage domain is shown to be shifted to a lower range of voltages, such as a range in which DAC feedback circuit **108** and comparator **102** are configured to operate.

[0090] When moving from voltage-shift phase **410-2** to conversion phase **410-3**, FIG. **5**C illustrated that new connections were made between input stage **106** and the downstream circuitry of DAC feedback circuit **108** and comparator **102**. These connections, which in FIG. **5**C were made based on the closing of switches **110-2**, **110-4**, and **110-6**, are represented in FIG. **6** as connections **610** on inputs **104-1** and **104-2**. As shown in the key below the circuit diagram in FIG. **6**, connections **610** will be understood to be disconnected during sampling phase **410-1** and voltage-shift phase **410-2** (illustrated by an open switch) and to be connected during conversion phase **410-3** (illustrated by a closed switch).

[0091] When connections **610** are made at the commencement of conversion phase **410-3**, graph **606** shows that waveforms **608-2** and **608-3** merge to form a single circuit node (the same node as input **104-2**). As shown, waveform **608-2** drops and waveform **608-3** raises to meet closer to a midscale voltage **612**. Once combined at the start of conversion phase **410-3**, graph **606** illustrates these two waveforms **608-2** and **608-3** as one waveform **608-23** that is drawn with a thicker line throughout the rest of conversion phase **410-3**. As described above, the connection of nodes **206-2** and **206-3** at the start of conversion phase **410-3** does not cause waveforms **608-2** and **608-3** to meet at a midscale voltage **612** due to charge held by transistors **202**. Instead, as shown, these devices cause waveforms **608-1** and **608-23** to form a differential signal around midscale voltage **612**. A differential voltage **614** between inputs **104-1** and **104-2** is therefore shown to be present after this point.

[0092] With MOS capacitance **602** and its sampled charge **604** now connected to a successive-approximation capacitance **616** (provided by the various successive-approximation capacitors **302**, as described above), comparator **102** and controller **112** may be configured to try to reduce differential voltage **614** as much as possible, thereby bringing waveforms **608-1** and **608-23** closer and closer to midscale voltage **612** with the addition of each new capacitor stage. As mentioned above, this is done by using comparator **102** to determine which polarity (positive or negative) is to be employed for each precharged capacitor in the series of successive-approximation capacitors. Accordingly, as shown in graph **606**, the differential signal created by waveforms **608-1** and **608-23** grows closer and closer to midscale voltage **612** in steps that each represent an addition of another properly aligned successive-approximation capacitor **302** to the overall successive-approximation capacitance **616**. As matched charge **618** gets close to actually matching sampled charge **604**, this differential signal approaches midscale voltage **612** and controller **112** may determine (based on the polarities that have been used to accomplish this) a digital value corresponding to the sampled input.

[0093] As has been mentioned, an additional benefit may arise from the use of transistor-based capacitance (such as MOS capacitance **602**) rather than a permanent type of capacitance that would be provided by an ordinary capacitor. Specifically, if the set of switches is manipulated to put the analog-to-digital converter circuit in the conversion configuration during a conversion phase, then: 1) at least one of the first transistor or the second transistor may be activated at a first time during the conversion phase, such that MOS capacitance **602** is enabled for use in storing the sampled charge **604** that was captured during a sampling phase prior to the first time; and 2) both the first transistor and the second transistor may be deactivated at a second time during the conversion phase, such that MOS capacitance **602** is disabled and a corresponding voltage boost is applied between the first output node (now connected to input **104-1**) and the second output node (now connected to input **104-2**) as sampled charge **604** is conserved. In other words, for example, as conversion phase **410-3** proceeds and differential voltage **614** decreases around midscale voltage **612**, there may become a point where both transistor **202-1**P and transistor **202-2**N are deactivated due to their threshold voltages no longer being met. At this point, the transistors shut off and the transistor capacitance they provide (MOS capacitance **602**) may be eliminated or significantly reduced. While the objective is for the controller to ultimately to cause the signal to decrease around the midscale voltage **612** as closely as possible, it will be understood that the differential voltage **614** may not decrease monotonically but may also have periods of temporary increase when certain new capacitors are added. Accordingly, transistors **202** may become deactivated and provide the voltage boost at one time during the conversion phase and become reactivated at a later time during the conversion phase.

[0094] Because charge is conserved, this reduction in capacitance in the circuit leads to a voltage boost in differential voltage **614** that is inversely proportional to the reduction in capacitance of MOS capacitance **602**, as modeled by the capacitance equation V=Q/C (where Q represents the charge being conserved, C represents the reduction in capacitance, and V represents the boost in voltage). Because ADC circuit **100** operates in the charge domain rather than the voltage domain, the total amount of charge being progressively distributed between MOS capacitance **602** and successive-approximation capacitance **616** is unaffected by this voltage boost, while the increase on differential voltage **614** helps make ADC circuit **100** less sensitive to noise, such as electrical noise originating in comparator **102**, as differential voltage **614** becomes more and more attenuated toward the latter part of conversion phase **410-3**.

[0095] Once matched charge **618** approximates sampled charge **604** to a suitable degree (e.g., based on the number of successive-approximation capacitors in the series, the amount of noise, etc.), the polarities of the successive-approximation capacitors may be used to represent a digital value corresponding to the original input signal. As such, the controller may output this digital value as digital output **120** and proceed back to the beginning of analog-to-digital conversion procedure **408** with a sampling phase **410-1** for the next value (not explicitly shown in FIG. **6**).

[0096] An analog-to-digital converter circuit with a single-ended input stage with CMOS-based capacitance, such as any of the implementations of ADC circuit **100** that have been described and illustrated herein, may be used in a variety of applications and use cases where analog-to-digital conversion is performed. As one particular example, such analog-to-digital converter circuits may be implemented as part of an image readout circuit included within an image sensor integrated circuit (IC). For example, an input stage of the analog-to-digital converter circuit may be associated with a particular column of a plurality of columns within the image sensor IC, and the input stage may receive, at the input node, a single-ended column readout signal for the particular column. In some examples, the manipulating of the set of switches to put the analog-to-digital converter circuit in the sampling configuration and/or in the conversion configuration are performed as steps of an image readout procedure performed by the image readout circuit.

[0097] To illustrate, FIG. **7**A shows certain aspects of an illustrative image sensor integrated circuit (IC) **700** within which one or more implementations of analog-to-digital converter circuitry described herein (such as implementations of ADC circuit **100**) may be included in accordance with principles described herein. It will be understood that, while certain implementations of analog-to-digital converter circuits described herein may relate to image sensors such as implemented by image sensor IC **700**, other implementations may be practiced in other contexts and/or without some or all of these particular details.

[0098] As shown in FIG. **7**A, image sensor IC **700** may include a pixel array **702** having multiple pixels **704** (sometimes referred to herein as image pixels **704** or image sensor pixels **704**) and row control circuitry **706** that is coupled to image pixel array **702**. Row control circuitry **706** may provide pixel control signals (e.g., row select signals, pixel reset signals, charge transfer signals, etc.) to pixels **704** over corresponding row control lines **708** to control the capture and read out of images using image sensor pixels in pixel array **702**.

[0099] Image sensor IC **700** may include column readout circuitry **710** and control and processing circuitry **712** that is coupled to row control circuitry **706** and column readout circuitry **710**. Column readout circuitry **710** may be coupled to pixel array **702** via multiple column lines **714**. For example, each column of pixels **704** in pixel array **702** may be coupled to a respective column line **714**. A corresponding analog-to-digital converter circuit **100** and (optional) column amplifier **716** may be interposed on each column line **714** for amplifying analog signals captured by pixel array **702** and converting the captured analog signals to corresponding digital pixel data. Column readout circuitry **710** may be coupled to external hardware such as control and processing circuitry **712**. Column readout circuitry **710** may perform column readout based on signals received from control and processing circuitry **712**. Column readout circuitry **710** may incorporate or be communicatively coupled to column ADC circuits **100** and/or column amplifiers **716**.

[0100] Column amplifiers **716** may be configured to receive analog signals (e.g., analog reset or image level signals) from pixel array **702** and to amplify the analog signals as may serve a particular implementation. Due to various features described herein for single-ended input stages with transistor-based capacitance that may be incorporated in ADC circuit **100**, column amplifiers **716** may be dispensed with or simplified in certain implementations. For example, certain tasks that might normally be performed by a pre-amplifier stage such as provided by column amplifiers **716** may be performed within ADC circuits **100**, such as by input stages described herein. As a few examples that have been described, the input stage may be responsible for performing a voltage shift and/or a single-ended-to-differential conversion for the analog signals. The analog signals may include data from a single column of pixels or from multiple columns of pixels, depending on the application. ADC circuits **100** may receive amplified analog signals from amplifiers **716** and may perform analog-to-digital conversion operations on the analog signals to generate digital data. The digital data may be transmitted to column readout circuitry **710** for processing and readout.

[0101] Pixel array **702** may have any number of rows and columns. In general, the size of pixel array **702** and the number of rows and columns in pixel array **702** will depend on the particular implementation of image sensor IC **700**. While rows and columns are generally described herein as being horizontal and vertical, respectively, rows and columns may refer to any grid-like structure (e.g., features described herein as rows may be arranged vertically and features described herein as columns may be arranged horizontally).

[0102] In FIG. **7**A, each column is illustrated as including a dedicated column line **714**, a dedicated column amplifier **716**, and a dedicated ADC circuit **100** for that particular column. Various advantages may arise from each column having dedicated resources in accordance with this type of implementation, when such resources are available. In certain implementations in which area or other resources are budgeted tightly, however, it may be desirable for multiple columns to share access to certain resources, such as certain aspects of the ADC circuit **100**. For example, a particular analog-to-digital converter circuit may implement a plurality of input stages that share parallel access to a singular comparator (comparator **102**) and a singular digital-to-analog feedback circuit (DAC feedback circuit **108**) of the analog-to-digital converter circuit. In this type of implementation, sampling and conversion operations may be staggered or pipelined across columns so as to share these resources more efficiently and thereby conserve area within image sensor IC **700**.

[0103] To illustrate, FIGS. **7**B and **7**C show certain aspects of an illustrative analog-to-digital converter circuit **720** that will be understood to be an implementation of ADC circuit **100** that includes a plurality of input stages **106-1** through **106**-N (where N is any suitable integer **2** or greater) that each receive respective analog signals through their own column lines **714-1** through **714**-N and (optional) column amplifiers **716-1** through **716**-N. As shown in FIG. **7**B, this plurality of input stages **106-1** through **106**-N may share access to a singular DAC feedback circuit **108** and a singular comparator **102** in this type of implementation. FIG. **7**C shows an example schematic diagram of illustrative analog-to-digital converter circuit **720** that includes the same inputs, outputs, and components (e.g., the set of switches, transistors, successive-approximation capacitors, etc.) as have been described in other examples above. In this example, the controller may be configured to further drive the switches to pipeline the various phases of analog-to-digital conversion procedure

408 such that the plurality of input stages **106** may share use of the DAC feedback circuit **108** and the comparator **102**.

[0104] A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the specification.

[0105] It will also be understood that when an element is referred to as being on, connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly on, connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly on, directly connected to or directly coupled to another element, there are no intervening elements present. Although the terms directly on, directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly on, directly connected or directly coupled can be referred to as such. The claims of the application may be amended to recite illustrative relationships described in the specification or shown in the figures.

[0106] The various apparatus and techniques described herein may be implemented using various semiconductor processing and/or packaging techniques. Some embodiments may be implemented using various types of semiconductor processing technologies associated with semiconductor substrates including, but not limited to, for example, Silicon (Si), Gallium Arsenide (GaAs), Silicon Carbide (SiC), and/or so forth.

[0107] As used in this specification, a singular form may, unless definitely indicating a particular case in terms of the context, include a plural form. Spatially relative terms (e.g., over, above, upper, under, beneath, below, lower, and so forth) are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. In some implementations, the relative terms above and below can, respectively, include vertically above and vertically below. In some implementations, the term adjacent can include laterally adjacent to or horizontally adjacent to.

[0108] While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the implementations. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The implementations described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different implementations described.

[0109] In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other embodiments are within the scope of the following claims.

[0110] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. A first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the implementations of the disclosure. As used herein, the term and/or includes any and all combinations of one or more of the associated listed items.

[0111] While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes, and equivalents may occur to those skilled in the art. It is therefore to be understood that the appended claims are intended to cover such modifications and changes as fall within the scope of the implementations. It will be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The implementations described herein can include various combinations and/or sub-combinations of the functions, components, and/or features of the different implementations described. As such, the scope of the present disclosure is not limited to the particular combinations hereafter claimed, but instead extends to encompass any combination of features or example implementations described herein irrespective of whether or not that particular combination has been specifically enumerated in the accompanying claims at this time.

Claims

1. An input stage for an analog-to-digital converter circuit, the input stage comprising: a first transistor of a first conductivity type, the first transistor being connected between a first node and a second node; a second transistor of a second conductivity type, the second transistor being connected between the first node and a third node; and a set of switches configured, when manipulated by a controller, to connect: the first node to either an input node or a first output node, the second node to either a first reference node or a second output node, and the third node to either a second reference node or the second output node.

2. The input stage of claim 1, wherein: the first conductivity type is a P-type and the second conductivity type is an N-type; the first transistor is connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node; and the second transistor is connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node.

3. The input stage of claim 1, wherein: the first conductivity type and the second conductivity type are both P-type; the first transistor is connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node; and the second transistor is connected between the first node and the third node by a second source and a second drain of the second transistor both connecting to the first node and a second gate of the second transistor connecting to the third node.

4. The input stage of claim 1, wherein: the first conductivity type is an N-type and the second conductivity type is a P-type; the first transistor is connected between the first node and the second node by a first source and a first drain of the first transistor both connecting to the first node and a first gate of the first transistor connecting to the second node; and the second transistor is connected between the first node and the third node by a second source and a second drain of the second transistor both connecting to the first node and a second gate of the second transistor connecting to the third node.

5. The input stage of claim 1, wherein: the first conductivity type and the second conductivity type are both an N-type; the first transistor is connected between the first node and the second node by a first source and a first drain of the first transistor both connecting to the first node and a first gate of the first transistor connecting to the second node; and the second transistor is connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node.

6. The input stage of claim 1, wherein the input node is configured to receive a single-ended signal as an input to the analog-to-digital converter circuit.

7. The input stage of claim 1, wherein the set of switches is manipulated to put the analog-to-digital converter circuit in a sampling configuration in which: the first node is connected to the input node

and disconnected from the first output node; the second node is connected to the first reference node and disconnected from the second output node; and the third node is connected to the second reference node and disconnected from the second output node.
8. The input stage of claim 1, wherein the set of switches is manipulated to put the analog-to-digital converter circuit in a conversion configuration in which: the first node is connected to the first output node and disconnected from the input node; the second node is connected to the second output node and disconnected from the first reference node; and the third node is connected to the second output node and disconnected from the second reference node.
9. The input stage of claim 8, wherein: the set of switches is manipulated to put the analog-to-digital converter circuit in the conversion configuration during a conversion phase; at a first time during the conversion phase, at least one of the first transistor or the second transistor is activated such that a transistor capacitance is enabled for use in storing a sampled charge that was captured during a sampling phase prior to the first time; and at a second time during the conversion phase, both the first transistor and the second transistor are deactivated such that the transistor capacitance is disabled and a corresponding voltage boost is applied between the first output node and the second output node as the sampled charge is conserved.
10. The input stage of claim 8, wherein: based on a sampled charge conserved within a transistor capacitance between the first output node and the second output node when the analog-to-digital converter circuit is in the conversion configuration, a differential signal corresponding to the sampled charge is generated on the first output node and the second output node; and the differential signal is received by a first input and a second input of a comparator included within the analog-to-digital converter circuit.
11. The input stage of claim 1, wherein the analog-to-digital converter circuit implements a charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converter that operates in a charge domain instead of a voltage domain.
12. The input stage of claim 1, wherein: the first transistor has a first bulk terminal and the second transistor has a second bulk terminal; and the set of switches is further configured, when manipulated by the controller, to connect: the first bulk terminal to either the first reference node or a first shifted supply node, and the second bulk terminal to either the second reference node or a second shifted supply node.
13. The input stage of claim 12, wherein the set of switches is manipulated to put the analog-to-digital converter circuit in a voltage-shift configuration in which: the first bulk terminal is connected to the first shifted supply node; the second bulk terminal is connected to the second shifted supply node; the first node is floating disconnected from both the input node and the first output node; the second node is floating disconnected from both the first reference node and the second output node; and the third node is floating disconnected from both the second reference node and the second output node.
14. The input stage of claim 1, wherein the input stage is one of a plurality of input stages that share parallel access to a singular comparator and a singular digital-to-analog feedback circuit of the analog-to-digital converter circuit.
15. The input stage of claim 1, wherein: the analog-to-digital converter circuit is implemented as part of an image readout circuit included within an image sensor integrated circuit; the input stage is associated with a particular column of a plurality of columns within the image sensor integrated circuit; and the input stage receives, at the input node, a single-ended column readout signal for the particular column.
16. An analog-to-digital converter circuit comprising: a comparator having a first input and a second input; an input stage including: a first transistor of a first conductivity type, the first transistor being connected between a first node and a second node; and a second transistor of a second conductivity type, the second transistor being connected between the first node and a third node; a digital-to-analog feedback circuit including a series of successive-approximation

capacitors; and a set of switches configured, when manipulated by a controller, to connect: the first node to either an input node or the first input of the comparator, the second node to either a first reference node or the second input of the comparator, the third node to either a second reference node or the second input of the comparator, and with either a positive polarity or a negative polarity, each capacitor of the series of successive-approximation capacitors between the first input and the second input of the comparator.

17. The analog-to-digital converter circuit of claim 16, wherein: the first conductivity type is a P-type and the second conductivity type is an N-type; the first transistor is connected between the first node and the second node by a first gate of the first transistor connecting to the first node and a first source and a first drain of the first transistor both connecting to the second node; and the second transistor is connected between the first node and the third node by a second gate of the second transistor connecting to the first node and a second source and a second drain of the second transistor both connecting to the third node.

18. The analog-to-digital converter circuit of claim 16, wherein: the input node is configured to receive a single-ended signal as an input to the analog-to-digital converter circuit; and the analog-to-digital converter circuit implements a charge-sharing successive-approximation-register (CS-SAR) analog-to-digital converter that operates in a charge domain instead of a voltage domain.

19. The analog-to-digital converter circuit of claim 16, wherein the set of switches is further configured, when manipulated by the controller, to connect each capacitor of the series of successive approximation capacitors between a first precharge reference node and a second precharge reference node.

20. A method for an analog-to-digital converter circuit that includes an input stage with a first transistor of a first conductivity type connected between a first node and a second node and a second transistor of a second conductivity type connected between the first node and a third node, the method comprising: manipulating a set of switches to put the analog-to-digital converter circuit in a sampling configuration in which: the first node is connected to an input node and disconnected from a comparator, the second node is connected to a first reference node and disconnected from the comparator, and the third node is connected to a second reference node and disconnected from the comparator; and manipulating the set of switches to put the analog-to-digital converter circuit in a conversion configuration in which: the first node is disconnected from the input node and connected to a first input of the comparator, the second node is disconnected from the first reference node and connected to a second input of the comparator, and the third node is disconnected from the second reference node and connected to the second input of the comparator.